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ORIGINAL ARTICLE

The 2021 US Preventive Services Task Force lung cancer screening eligibility criteria disproportionately exclude younger Black patients with lung cancer

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Abstract

Background: The 2021 US Preventive Services Task Force (USPSTF) lung cancer screening guideline may continue to exclude many younger Black individuals who have not yet accumulated enough smoking pack-years to be eligible for screening. The objective of this study was to evaluate the proportions of Black and White patients with lung cancer, stratified by age at diagnosis, who would have been eligible for lung cancer screening.

Methods: Incident lung cancer cases among Black and White individuals aged 50–80 years with a smoking history in the Southern Community Cohort Study (SCCS) were identified for analysis. The proportions of Black and White individuals, stratified by age at diagnosis, who would have qualified for screening under the 2013 and 2021 USPSTF guidelines were calculated.

Results: A total of 1856 individuals met inclusion criteria. Compared to the 2013 USPSTF guideline, the 2021 USPSTF guideline significantly increased the proportions of Black and White patients with lung cancer who would have qualified for screening in all age groups evaluated. However, under the 2021 USPSTF guideline, there remained a notable racial disparity in lung cancer screening eligibility, particularly among younger patients with lung cancer. Only 47.4% and 61.9% of Black patients aged 50–54 and 50–59 years, respectively, would have qualified for screening under the 2021 USPSTF guideline compared to 80.3% and 88.8% of White patients aged 50–54 and 50–59 years, respectively. With increasing age, the racial disparity in lung cancer screening eligibility between Black and White patients with lung cancer was reduced, and there were no longer any statistically significant differences in screening eligibility between Black and White patients with lung cancer aged 70–74 and 75–80 years. Of Black individuals aged 50–54 and 55–59 years who were ineligible for screening, 100% and 97.5% were ineligible because they had smoked fewer than 20 pack-years, respectively.

Alexandra L. Potter and Sangkavi Kuhan contributed equally to this article.

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Conclusions: This analysis of SCCS participants with lung cancer found that the 2021 USPSTF eligibility criteria disproportionately exclude many younger Black individuals with lung cancer, primarily because they have too few smoking pack-years.

KEYWORDS
early detection, lung cancer, lung cancer screening eligibility, pack-years, racial disparities

INTRODUCTION

Lung cancer screening with low-dose computed tomography was first recommended by the US Preventive Services Task Force (USPSTF) in 2013 for individuals aged 55–80 years who have smoked at least 30 pack-years and who either currently smoke or quit smoking within the past 15 years.[1] However, these eligibility criteria were found to exclude many high-risk individuals, especially those from racial and ethnic minority groups. A previous analysis of the Southern Community Cohort Study (SCCS) by Aldrich and colleagues[2] found that only 32% of Black patients with lung cancer would have qualified for screening compared to 56% of White patients with lung cancer under the 2013 USPSTF guideline; this disparity arose as a result of Black individuals being both younger and having lighter smoking histories at lung cancer diagnosis when compared to White individuals. In 2021, to address race- and sex-based disparities in screening eligibility, the USPSTF updated its lung cancer eligibility criteria by decreasing the lower age bound from 55 to 50 years and the pack-year smoking history requirement from 30 to 20 pack-years.[3]

However, the 2021 USPSTF eligibility criteria may continue to exclude many younger (e.g., aged 50–59 years) Black individuals at high risk for lung cancer who—despite meeting the 2021 USPSTF age requirements—have not yet accumulated enough smoking pack-years to be eligible for screening. The objective of this study was to evaluate the proportions of Black and White patients with lung cancer, stratified by age at diagnosis, who would have been eligible for lung cancer screening in the SCCS. We hypothesized that many younger (i.e., aged 50–59 years) Black individuals diagnosed with lung cancer had smoked fewer than 20 pack-years at the time of lung cancer diagnosis, and thus still would not have qualified for screening under the 2021 USPSTF lung cancer screening guideline.

MATERIALS AND METHODS

Data source: The SCCS

The SCCS is a large prospective cohort study that enrolled approximately 85,000 predominantly low-income Black and White adults between March 25, 2002, and September 24, 2009, from 12 southeastern states in the United States (Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, and West Virginia).[4] SCCS participants completed a baseline questionnaire at enrollment and up to three additional questionnaires, administered every 5 years, during follow-up. Each questionnaire collected detailed information on cigarette smoking characteristics (i.e., smoking status, pack-years, duration, and intensity). Incident lung cancer cases diagnosed among individuals in the SCCS were collected via linkage with state cancer registries up to December 31, 2021. Informed consent was obtained from each individual participating in the SCCS. The SCCS was approved by the institutional review boards at Vanderbilt University and Meharry Medical College, both in Nashville, Tennessee.

Study design

This study was approved by the Mass General Brigham Institutional Review Board. Black and White SCCS participants aged 50–80 years with a history of smoking who were diagnosed with lung cancer from 2002 to 2021 were identified for analysis. SCCS participants diagnosed with lung cancer who had never smoked, had missing smoking information, or had a history of lung cancer at the time of enrollment into the SCCS were excluded from this analysis (Figure 1), which resulted in a final sample of 1856 participants. Information regarding participants' smoking history (i.e., smoking status, duration, intensity, and pack-year smoking history) was collected from the baseline and follow-up questionnaires. Smoking history was first collected from the baseline questionnaire and then updated according to data from follow-up questionnaires (such that the smoking history recorded for a given patient reflected that patient's smoking status and history at the time closest to their lung cancer diagnosis) (see Supporting Information S1: Supplementary Appendix for further details).

SCCS participants diagnosed with lung cancer were grouped according to their age at lung cancer diagnosis: 50–54, 55–59, 60–64, 65–69, 70–74, and 75–80 years. Differences in smoking characteristics between Black and White patients with lung cancer in each age group were evaluated. The proportions of Black and White patients with lung cancer in each age group who smoked half a pack of cigarettes per day or less were examined. Among Black and White patients with lung cancer who smoked half a pack of cigarettes per day or less, the relationship between age at lung cancer diagnosis and pack-year smoking history was evaluated.

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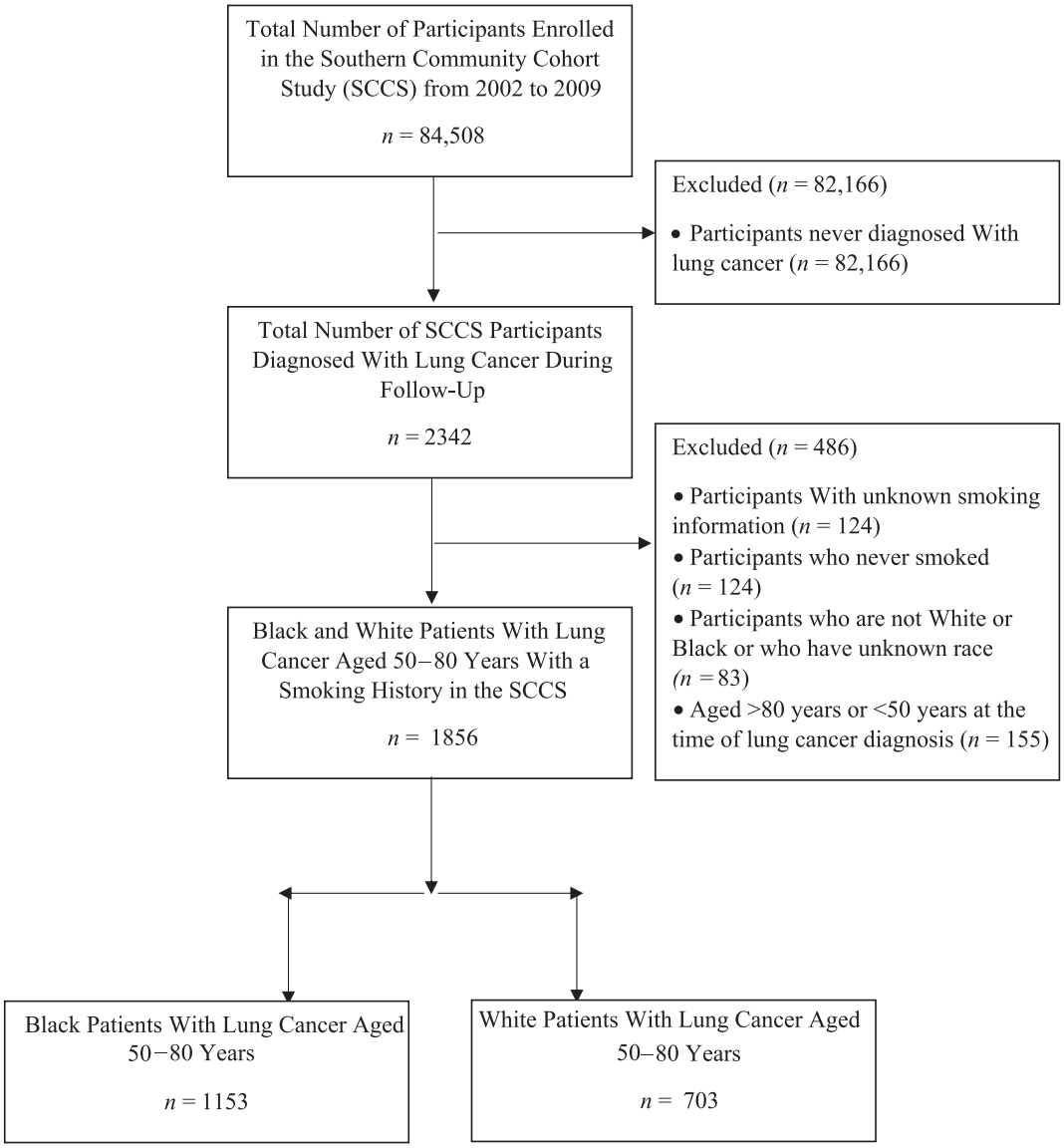


FIGURE 1 Consolidated Standards of Reporting Trials diagram.

Next, the proportions of Black and White patients with lung cancer in each age group who would have qualified for screening under the 2013 and 2021 USPSTF guidelines were evaluated. Reasons for ineligibility under the 2021 USPSTF guideline were examined among Black and White participants diagnosed with lung cancer, stratified by age group.

Statistical analysis

Changes in the proportion of Black and White patients with lung cancer eligible for screening under the 2013 versus 2021 USPSTF guidelines were evaluated with the McNemar test. Differences in the proportion of Black versus White patients with lung cancer eligible for screening under the 2013 and 2021 USPSTF guidelines were evaluated with the Pearson χ^2 test.

All statistical analyses were performed with STATA, version 17.0. p values were two sided and considered statistically significant if $p < .05$.

RESULTS

A total of 1856 individuals diagnosed with lung cancer in the SCCS met study inclusion criteria, of which 703 (37.9%) were White and 1153 (62.1%) were Black (Figure 1). Of these participants, 225 (12.1%) were aged 50–54 years, 428 (23.1%) were aged 55–59 years, 409 (22.0%) were aged 60–64 years, 351 (18.9%) were aged 65–69 years, 278 (15.0%) were aged 70–74 years, and 165 (8.9%) were aged 75–80 years. Baseline characteristics of Black and White SCCS participants diagnosed with lung cancer, stratified by age group, are shown in Table 1.

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TABLE 1 Baseline characteristics of Black and White patients with lung cancer in the Southern Community Cohort Study stratified by age at lung cancer diagnosis.

	Age 50–54 years		Age 55–59 years		Age 60–64 years		Age 65–69 years		Age 70–74 years		Age 75–80 years	
Characteristic	**White**	**Black**	**White**	**Black**	**White**	**Black**	**White**	**Black**	**White**	**Black**	**White**	**Black**
Patients, No.	71	154	116	312	158	251	152	199	134	144	72	93
Sex, No. (%)												
Female	45 (63.4)	66 (42.9)	71 (61.2)	123 (39.4)	90 (57.0)	102 (40.6)	85 (55.9)	82 (41.2)	77 (57.5)	68 (47.2)	37 (51.4)	44 (47.3)
Male	26 (36.6)	88 (57.1)	45 (38.8)	189 (60.6)	68 (43.0)	149 (59.4)	67 (44.1)	117 (58.8)	57 (42.5)	76 (52.8)	35 (48.6)	49 (52.7)
Age at diagnosis, median (IQR), years	53.0 (52.0–54.0)	52.0 (51.0–54.0)	58.0 (56.0–58.0)	57.0 (56.0–58.0)	62.0 (61.0–63.0)	62.0 (61.0–63.0)	67.0 (66.0–68.0)	67.0 (65.0–68.0)	72.0 (70.0–73.0)	71.0 (70.0–73.0)	77.0 (76.0–79.0)	77.0 (75.0–78.0)
Smoking status at diagnosis, No. (%)												
Former	8 (11.3)	22 (14.3)	16 (13.8)	39 (12.5)	31 (19.6)	50 (19.9)	56 (36.8)	52 (26.1)	62 (46.3)	52 (36.1)	51 (70.8)	52 (55.9)
Current	63 (88.7)	132 (85.7)	100 (86.2)	273 (87.5)	127 (80.4)	201 (80.1)	96 (63.2)	147 (73.9)	72 (53.7)	92 (63.9)	21 (29.2)	41 (44.1)
Smoking history, median (IQR), pack-years	41.0 (28.0–57.0)	19.0 (14.4–35.1)	44.0 (31.1–63.8)	22.5 (17.5–40.0)	48.7 (39.0–69.0)	26.0 (17.5–47.0)	53.0 (34.8–76.1)	27.0 (17.5–47.0)	52.3 (29.0–77.3)	27.5 (15.4–49.8)	59.3 (32.5–86.3)	30.0 (16.8–53.0)
Pack-year category, No. (%)												
≤10 pack-years	4 (5.6)	21 (13.6)	4 (3.4)	34 (10.9)	3 (1.9)	36 (14.3)	7 (4.6)	19 (9.5)	9 (6.7)	17 (11.8)	5 (6.9)	10 (10.8)
11–19 pack-years	10 (14.1)	60 (39.0)	9 (7.8)	82 (26.3)	8 (5.1)	36 (14.3)	10 (6.6)	38 (19.1)	11 (8.2)	25 (17.4)	6 (8.3)	17 (18.3)
20–29 pack-years	4 (5.6)	22 (14.3)	12 (10.3)	69 (22.1)	17 (10.8)	62 (24.7)	16 (10.5)	50 (25.1)	17 (12.7)	38 (26.4)	6 (8.3)	19 (20.4)
≥30 pack-years	53 (74.6)	51 (33.1)	91 (78.4)	127 (40.7)	130 (82.3)	117 (46.6)	119 (78.3)	92 (46.2)	97 (72.4)	64 (44.4)	55 (76.4)	47 (50.5)
Cigarettes per day at baseline, median (IQR)	20.0 (15.0–30.0)	12.0 (10.0–20.0)	20.0 (15.0–30.0)	12.0 (10.0–20.0)	20.0 (20.0–30.0)	12.0 (10.0–20.0)	20.0 (20.0–30.0)	12.0 (8.0–20.0)	20.0 (15.0–30.0)	10.0 (7.0–20.0)	20.0 (20.0–40.0)	10.0 (9.0–20.0)
Smoking duration, median (IQR), years	38.5 (35.0–41.0)	35.0 (31.5–38.0)	42.0 (39.3–44.0)	40.0 (37.0–43.0)	46.0 (42.0–49.0)	44.0 (40.0–47.0)	49.3 (42.0–53.5)	47.0 (43.0–52.0)	52.0 (43.0–56.0)	52.0 (47.0–56.0)	49.0 (37.5–59.0)	52.0 (43.0–60.0)
Time since quitting smoking, median (IQR), years	4.8 (2.8–7.0)	5.0 (3.0–8.0)	7.0 (3.3–8.5)	5.0 (3.0–11.0)	8.0 (5.5–15.0)	7.5 (4.0–16.0)	8.0 (4.0–14.0)	9.0 (5.0–17.0)	11.0 (6.0–18.0)	10.5 (6.3–19.5)	16.0 (9.0–26.0)	12.3 (7.3–28.0)

Abbreviation: IQR, interquartile range.

Across all age groups, Black patients with lung cancer smoked significantly fewer cigarettes per day compared to White patients with lung cancer, with 46%–56% of Black patients with lung cancer smoking half a pack of cigarettes per day or less compared to 13%–23% of White patients with lung cancer (Figure 2).

Figure 3 shows the distribution of age at diagnosis and pack-year smoking history among Black and White patients with lung cancer who smoked half a pack of cigarettes per day or less. Notably, the large majority (81.3%) of patients with lung cancer who smoked half a pack of cigarettes per day or less were Black. Among Black and White patients with lung cancer who smoked half a pack of cigarettes per day or less, pack-year smoking history tended to increase with increasing age. Between 40% and 50% of patients aged older than 60 years at the time of lung cancer diagnosis who smoked half a pack of cigarettes per day or less had accumulated at least 20 pack-years by the time of lung cancer diagnosis. However, only 2.2% of patients aged 50–54 years and 27.4% of patients aged 55–59 years at the time of lung cancer diagnosis who smoked half a pack of cigarettes per day or less had accumulated 20 pack-years.

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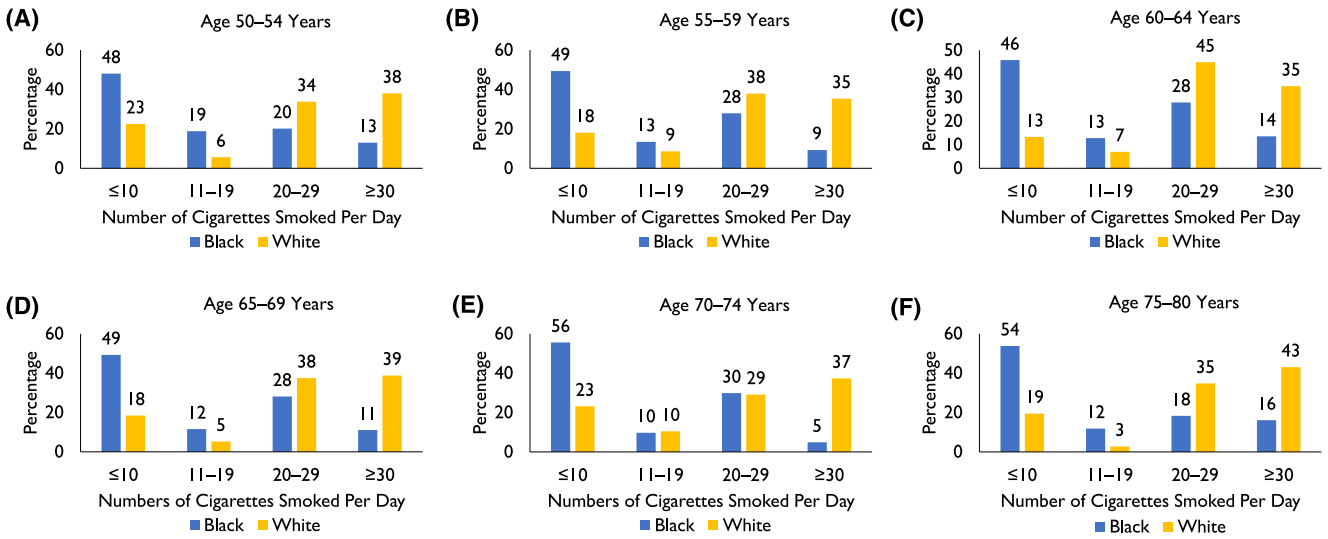


FIGURE 2 Distribution of smoking intensity (cigarettes per day) among Black and White lung cancer patients diagnosed with lung cancer stratified by age group: age 50–54 (A), 55–59 (B), 60–64 (C), 65–69 (D), 70–74 (E), and 75–80 (F).

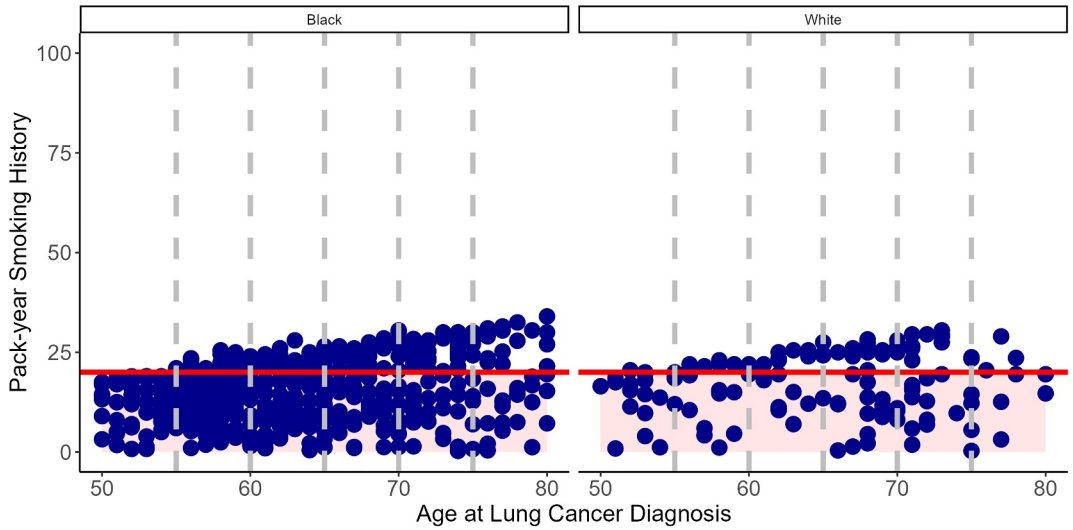


FIGURE 3 Distribution of age at lung cancer diagnosis versus pack-year smoking history among Black and White patients with lung cancer who smoked half a pack of cigarettes per day or less. Each dark blue dot represents a patient with lung cancer. The red solid line indicates the 20-pack-year smoking history cutoff. The gray dashed lines indicate the different age group cutoffs.

Eligibility under the 2013 USPSTF guideline stratified by age group

The percentage of Black and White patients with lung cancer who would have been eligible for screening under the 2013 USPSTF guideline, stratified by age group, is shown in Figure 4A. Under the 2013 USPSTF guideline, 40.4% of Black patients aged 55–59 years would have qualified for screening compared to 78.5% of White patients aged 55–59 years ($p < .001$). Similar differences in screening eligibility under the 2013 USPSTF guideline were observed among Black and White patients with lung cancer aged 60–64 ($p < .001$), 65–69 ($p < .001$), and 70–74 years ($p < .001$). Among individuals aged 75–80 years, a lower proportion of Black patients with lung cancer would have qualified for screening under the 2013 USPSTF guideline compared to White patients with lung cancer but this difference was not statistically significant (38.7% vs. 50.0%; $p = .15$).

Eligibility under the 2021 USPSTF guideline stratified by age group

The percentage of Black and White patients with lung cancer who would have been eligible for screening under the 2021 USPSTF guideline, stratified by age group, is shown in Figure 4B. The changes made in the 2021 USPSTF guideline led to significant increases in eligibility for both Black and White individuals in all age groups evaluated, with the largest increases occurring among Black individuals (Table 2). The disparity in eligibility between Black and

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White patients with lung cancer was reduced under the 2021 USPSTF guideline when compared to the 2013 USPSTF guideline in all age groups evaluated (Table 2). For example, under the 2013 USPSTF guideline, among individuals aged 55–59 years, there was a 38.1–percentage point difference (95% CI, 28.8–47.3 percentage points) in the percentage of Black versus White patients with lung cancer eligible for screening. Under the 2021 USPSTF guideline, this difference in the percentage of Black versus White patients with lung cancer aged 55–59 years eligible for screening was reduced to 26.9 percentage points (95% CI, 19.1–34.8 percentage points).

However, despite the increases in eligibility observed under the 2021 USPSTF guideline compared to the 2013 USPSTF guideline,

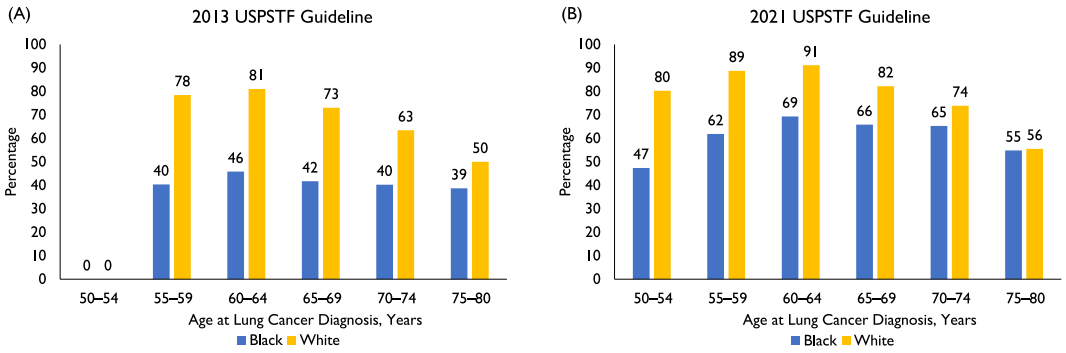


FIGURE 4 Eligibility of Black versus White patients with lung cancer for screening under the 2013 USPSTF guideline (A) and 2021 USPSTF guideline (B) stratified by age at lung cancer diagnosis. USPSTF indicates US Preventive Services Task Force.

TABLE 2 Percentage eligible, relative change in eligibility, and percentage point difference in eligibility between Black and White patients under the 2013 and 2021 USPSTF guidelines for lung cancer screening.

	Percentage eligible under the 2013 USPSTF guideline, % (*n* of *N*)	Percentage eligible under the 2021 USPSTF guideline, % (*n* of *N*)	Relative change in eligibility under the 2013 versus 2021 USPSTF guidelines, % (95% CI)	Percentage point difference in eligibility between Black and White patients under the 2013 USPSTF criteria, % (95% CI)	Percentage point difference in eligibility between Black and White patients under the 2021 USPSTF criteria, % (95% CI)
Age 50–54 years					
Black	0 (0 of 154)	47.4 (73 of 154)	NA	NA	32.9 (20.7 to 45.0)
White	0 (0 of 71)	80.3 (57 of 71)	NA		
Age 55–59 years					
Black	40.4 (126 of 312)	61.9 (193 of 312)	53.2 (38.2 to 69.7)	38.1 (28.8 to 47.3)	26.9 (19.1 to 34.8)
White	78.5 (91 of 116)	88.8 (103 of 116)	13.2 (5.5 to 21.4)		
Age 60–64 years					
Black	45.8 (115 of 251)	69.3 (174 of 251)	51.3 (36.0 to 68.3)	35.2 (26.5 to 43.9)	21.8 (14.6 to 29.0)
White	81.0 (128 of 158)	91.1 (144 of 158)	12.5 (6.2 to 19.2)		
Age 65–69 years					
Black	41.7 (83 of 199)	65.8 (131 of 199)	57.8 (38.6 to 79.8)	31.3 (21.5 to 41.2)	16.4 (7.4 to 25.4)
White	73.0 (111 of 152)	82.2 (125 of 152)	12.6 (5.8 to 19.8)		
Age 70–74 years					
Black	40.3 (58 of 144)	65.3 (94 of 144)	62.1 (38.2 to 90.1)	23.2 (11.7 to 34.6)	8.6 (−2.2 to 19.4)
White	63.4 (85 of 134)	73.9 (99 of 134)	16.5 (7.5 to 26.1)		
Age 75–80 years					
Black	38.7 (36 of 93)	54.8 (51 of 93)	41.2 (18.7 to 69.1)	11.3 (−3.9 to 26.5)	0.7 (−14.6 to 16.0)
White	50.0 (36 of 72)	55.6 (40 of 72)	11.1 (0.2 to 23.2)		

Abbreviations: NA, Not Applicable; USPSTF, US Preventive Services Task Force.

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there remained a notable disparity in screening eligibility between Black and White patients with lung cancer under the 2021 USPSTF guideline, particularly among younger individuals aged 50–54 and 55–59 years. Among individuals aged 50–54 years, only 47.4% of Black patients would have qualified compared to 80.3% of White patients ($p < .001$). Among individuals aged 55–59 years, 61.9% of Black patients would have qualified compared to 88.8% of White patients ($p < .001$).

With increasing age, the disparity in the proportion of Black versus White patients who would have qualified for lung cancer screening decreased (Figure 4B), although it remained statistically significant for individuals in all age groups of <70 years. Notably, there were no statistically significant differences in the proportion of Black versus White patients with lung cancer who would have qualified for lung cancer screening among individuals aged 70–74 years (65.3% vs. 73.9%; $p = .12$) and 75–80 years (54.8% vs. 55.6%; $p = .93$).

Reasons for ineligibility under the 2021 USPSTF guideline

Table 3 shows the reasons for ineligibility under the 2021 USPSTF guideline by race and age group. The 20–pack-year cutoff was the primary reason for ineligibility among Black and White patients with lung cancer aged 50–54, 55–59, 60–64, and 65–69 years. Notably, 100% of Black and White patients with lung cancer aged 50–54 years who were ineligible for screening under the 2021 USPSTF guideline had smoked fewer than 20 pack-years at lung cancer diagnosis. With increasing age, a greater proportion of Black and White patients with lung cancer were ineligible because they had quit smoking more than 15 years previously. However, the 20–pack-year cutoff remained the primary reason for ineligibility among Black patients with lung cancer in all age groups.

DISCUSSION

In this analysis of Black and White patients with lung cancer with a history of smoking in the SCCS, we found that the changes made to the 2021 USPSTF lung cancer screening guideline significantly increased the proportion of Black and White patients with lung cancer who would have qualified for screening in all age groups evaluated. However, despite the increases in eligibility observed under the 2021 USPSTF guideline, there remained a notable racial disparity in lung cancer screening eligibility between Black and White patients with lung cancer, particularly among younger patients aged 50–54 and 55–59 years. Notably, only 47.4% of Black patients aged

TABLE 3 Reasons for ineligibility among Black and White patients with lung cancer ineligible for lung cancer screening under the 2021 USPSTF guideline stratified by age group.

	Black, age 50–54 years, %	***n* of *N***	**White, age 50–54 years, %**	***n* of *N***
<20 pack-years	100	81 of 81	100	14 of 14
>15 years quit	3.70	3 of 81	0	0 of 14
	Black, age 55–59 years, %	***n* of *N***	**White, age 55–59 years, %**	***n* of *N***
<20 pack-years	97.50	116 of 119	100	13 of 13
>15 years quit	4.20	5 of 119	7.70	1 of 13
	Black, age 60–64 years, %	***n* of *N***	**White, age 60–64, years, %**	***n* of *N***
<20 pack-years	93.5	72 of 77	78.6	11 of 14
>15 years quit	16.9	13 of 77	50.0	7 of 14
	Black, age 65–69 years, %	***n* of *N***	**White, age 65–69 years, %**	***n* of *N***
<20 pack-years	83.2	57 of 68	63.0	17 of 27
>15 years quit	23.5	16 of 68	44.4	12 of 27
	Black, age 70–74 years, %	***n* of *N***	**White, age 70–74 years, %**	***n* of *N***
<20 pack-years	84.0	42 of 50	57.1	20 of 35
>15 years quit	32.0	16 of 50	57.1	20 of 35
	Black, age 75–80 years, %	***n* of *N***	**White, age 75–80 years, %**	***n* of *N***
<20 pack-years	64.3	27 of 42	34.4	11 of 32
>15 years quit	52.4	22 of 42	84.4	27 of 32

Note: Reasons for ineligibility are not mutually exclusive (i.e., an individual may have smoked fewer than 20 pack-years and also quit smoking more than 15 years ago).

Abbreviation: USPSTF, US Preventive Services Task Force.

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50–54 years would have qualified for screening under the 2021 USPSTF guideline compared to 80.3% of White patients aged 50–54 years. With increasing age, the racial disparity in lung cancer screening eligibility between Black and White patients with lung cancer was reduced, and there were no longer any statistically significant differences in lung cancer screening eligibility between Black and White patients with lung cancer aged 70–74 and 75–80 years.

Previous studies have reported that Black individuals diagnosed with lung cancer are younger and smoke less intensely compared to White individuals with lung cancer.[2,5,6] Indeed—in the present study—we found that approximately 50% of Black patients with lung cancer, across all age groups, smoked half a pack of cigarettes per day or less compared to less than 20% of White patients with lung cancer. To address these important differences in age, smoking behavior, and lung cancer risk between Black and White individuals, in 2021 the USPSTF updated its lung cancer screening guideline by decreasing the lower age bound from 55 to 50 years and the pack-year smoking requirement from 30 to 20 pack-years.[3] Use of a 20–pack-year cutoff as opposed to a 30–pack-year cutoff in the USPSTF guideline is based on data demonstrating that this lower pack-year cutoff increased the number of high-risk individuals eligible for screening and decreased race- and sex-based disparities in screening eligibility.[7] However, it is important to note that the 20–pack-year cutoff included in the 2021 USPSTF guideline does not necessarily mark the point at which lung cancer risk becomes sufficiently high to warrant lung cancer screening.

When we examined the relationship between age at lung cancer diagnosis and pack-year smoking history among individuals who smoke less intensely (i.e., half a pack of cigarettes per day or less), we found that the use of a 20–pack-year cutoff to determine screening eligibility excluded many of these individuals between the ages of 50 and 59 years. Given that individuals typically start smoking between the ages of 15 and 20 years,[8,9] most individuals will have accumulated 30–35 years of smoking by age 50 years. Among individuals who smoke half a pack of cigarettes per day or less, provided they continually smoke, these individuals will not meet the 20–pack-year requirement until at least age 55 years or beyond. Importantly, we found that only 2.2% of individuals aged 50–54 years and 27.4% of individuals aged 50–59 years diagnosed with lung cancer who smoked half a pack of cigarettes per day or less would have reached 20 pack-years by the time of their lung cancer diagnosis. Because individuals who smoke half a pack a day or less are disproportionately Black (81% of individuals in our cohort smoking half a pack per day or less were Black compared to 50% of individuals smoking more than half a pack per day), the 20–pack-year cutoff is a major driver of disparities in screening eligibility between Black and White individuals, and likely explains the large disparity in screening eligibility under the 2021 USPSTF guideline among Black and White patients with lung cancer aged 50–54 and 50–59 years in our cohort.

It is worth noting that there were marked differences in the sex distribution by race in the study cohort, with a higher proportion of patients who were female among the White patients compared to Black patients. Notably, the greatest sex imbalance occurred among patients with lung cancer in the youngest age groups (i.e., 50–54 and 55–59 years). For example, 63% of White patients aged 50–54 years were female compared to only 43% of Black patients aged 50–54 years. The differences in smoking history and lung cancer screening eligibility between Black and White patients with lung cancer observed in the present study may have been attributed, in part, to this sex imbalance. It is important to note, however, that female patients diagnosed with lung cancer have been shown to smoke less and to be less likely to be eligible for lung cancer screening compared to male patients diagnosed with lung cancer.[10,11] Therefore, a preponderance of female patients among the White patients with lung cancer would likely contribute to lighter smoking histories and lower lung cancer screening eligibility among the White patients with lung cancer (compared to if there had been a higher proportion of male White patients with lung cancer). If the distribution by sex was more balanced between the Black and White patients, it is likely that the smoking differences and disparities in screening eligibility by race would be even larger compared to that observed in the present study.

The findings of this study demonstrate the need for efforts to address the large racial disparity in screening eligibility under the 2021 USPSTF guideline between Black and White individuals, especially among those in younger age groups (i.e., individuals between the ages of 50 and 69 years). To date, several strategies have been proposed to address the persistent racial disparities in lung cancer screening eligibility under the 2021 USPSTF guideline. One proposed strategy to reduce racial disparities in screening eligibility is the use of risk prediction models, such as the prostate, lung, colorectal, and ovarian (PLCO) m2012 model[12] and the life-years from screening with computed tomography (LYFS-CT) model.[13] Several previous studies have reported that using the PLCOm2012 risk prediction model to select individuals for screening, instead of the categorical age and smoking criteria in the 2021 USPSTF guideline, reduces or eliminates racial disparities in screening eligibility.[14–16] Additionally, a study by Landy and colleagues reported that augmenting the 2021 USPSTF criteria with the LYFS-CT model notably reduced racial disparities in screening eligibility between Black and White individuals.[13] The findings from these previous studies support that using risk prediction models to select individuals for screening may present one strategy to help improve equity in opportunities for screening. However, it should be noted that the USPSTF has previously expressed concern that the added complexity of risk prediction models may pose a barrier to the widespread uptake of lung cancer screening.[17] Although the use of risk prediction models to select individuals for screening may, in theory, reduce disparities in screening eligibility, it is possible that challenges implementing these models—especially in lower resource settings, where individuals from racial and ethnic minority groups disproportionately reside—may mitigate their overall impact.

Another approach to reducing racial disparities in lung cancer screening eligibility is the use of a smoking duration threshold, instead of a smoking pack-year threshold, to select individuals for lung cancer screening.[18] We previously showed that replacing the 20–pack-year criterion in the USPSTF guideline with a 20-year smoking duration criterion eliminated the racial disparity in

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screening eligibility between Black and White individuals.[18] Notably, in our previous analysis,[18] we found that replacing the 20–pack-year requirement with a 20-year smoking duration requirement led to significant increases in eligibility among younger individuals, which suggests that using duration, instead of pack-years, may be one approach to eliminate the racial disparity between younger Black and White individuals in lung cancer screening eligibility. A considerable strength of this approach is that it is simple—even simpler than the current USPSTF guideline—because total smoking duration is easier to calculate when compared to pack-year smoking history.

There are important limitations to this study. First, this study is a descriptive analysis of age and smoking patterns among individuals diagnosed with lung cancer. It is important to note that the findings of the present study are limited by their descriptive nature. Second, the 70–74 and 75–80 year age groups evaluated in our analysis had smaller sample sizes, and we may have been underpowered to detect statistically significant differences in lung cancer screening eligibility between Black and White patients in these age groups. Third, smoking information is self-reported. As a result, it is possible that smoking information (e.g., pack-years, duration, and intensity) was misclassified. However, the SCCS has published numerous studies associating cigarette smoking with the risk of adverse health outcomes, with findings as expected from the literature.[19,20] Fourth, a small portion (~5%) of patients with lung cancer in the SCCS had unknown smoking information, and thus were not included in our analysis. Last, the SCCS participants diagnosed with lung cancer are not representative of all patients with lung cancer in the United States, and our findings may not be generalizable to populations outside of the SCCS. That said, a considerable strength of the SCCS is that it comprises a higher risk population (i.e., it is enriched for individuals who currently smoke) from 12 southeastern states, among whom it is important to study lung cancer screening eligibility.

In conclusion, in this descriptive analysis of Black and White patients with lung cancer with a history of smoking in the SCCS, we found that the changes made in the 2021 USPSTF lung cancer screening guideline significantly increased the proportion of Black and White patients with lung cancer who would have qualified for screening in all age groups evaluated. However, despite the increases in eligibility observed under the 2021 USPSTF guideline, there remained a notable racial disparity in lung cancer screening eligibility between Black and White patients with lung cancer, particularly among younger patients aged 50–54 and 55–59 years. Notably, only 47.4% of Black patients aged 50–54 years would have qualified for screening under the 2021 USPSTF guideline compared to 80.3% of White patients aged 50–54 years. The primary reason for ineligibility among Black patients with lung cancer was having fewer than 20 pack-years at the time of lung cancer diagnosis. Given that younger patients are often healthier and stand to benefit the most from early detection and treatment, this difference in screening eligibility among individuals aged 50–54 and 55–59 years may worsen existing disparities in the receipt of curative-intent treatment and survival between White and Black patients with lung cancer. These findings demonstrate the urgent need for efforts to address the large disparity in lung cancer screening eligibility between younger Black and White individuals under the 2021 USPSTF guideline. Revising the USPSTF criteria, such as by replacing the 20–pack-year requirement with a 20-year smoking duration requirement, and augmenting the USPSTF criteria with the results from risk prediction models, may present paths forward to improve the equity in opportunities for early lung cancer detection.

AUTHOR CONTRIBUTIONS

Alexandra L. Potter: Conceptualization, methodology, data curation, writing–original draft, writing–review and editing, formal analysis, visualization, and software. **Sangkavi Kuhan**: Conceptualization, methodology, data curation, formal analysis, writing–original draft, writing–review and editing, visualization, and software. **Priyanka Senthil**: Writing–original draft, writing–review and editing, visualization, formal analysis, and data curation. **Arian Mansur**: Writing–original draft and data curation. **Chinmay Haridas**: Data curation and writing–original draft. **Deepti Srinivasan**: Writing–original draft, formal analysis, and data curation. **Arvind Kumar**: Writing–original draft and formal analysis. **Wei Zheng**: Conceptualization, writing–original draft, formal analysis, and data curation. **Erica T. Warner**: Conceptualization, writing–original draft, formal analysis, and methodology. **Chi-Fu Jeffrey Yang**: Conceptualization, project administration, writing–original draft, writing–review and editing, formal analysis, data curation, and supervision.

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CONFLICT OF INTEREST STATEMENT

Alexandra L. Potter, Sangkavi Kuhan, Priyanka Senthil, Deepti Srinivasan, and Chi-Fu Jeffrey Yang report leadership roles in the American Lung Cancer Screening Initiative. Erica T. Warner has received grants from Pfizer and has consulting or advisory roles with AstraZeneca. Chi-Fu Jeffrey Yang has consulting or advisory roles with AstraZeneca and Genentech, has received honoraria from AstraZeneca and Genentech, and has received travel support from AstraZeneca. The other authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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